HIGH BEAM INDICATOR (Continued)

cator bulb and the cluster control circuitry by performing the instrument cluster actuator test. (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the high beam indicator fails to illuminate during the actuator test, replace the indicator bulb and bulb holder with a known good unit and repeat the test. If the indicator still fails to illuminate, replace the faulty instrument cluster. If the problem being diagnosed is related to inoperative headlamp high beams, be certain to repair the headlamp system circuits and switches before attempting to diagnose or repair the high beam indicator. If no headlamp system problems are found and the high beam indicator illuminates during the instrument cluster actuator test, the following procedure will help locate an open in the high beam indicator driver circuit. Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.

WARNING: TO AVOID PERSONAL INJURY OR DEATH, ON VEHICLES EQUIPPED WITH AIRBAGS, DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM BEFORE ATTEMPTING ANY STEERING WHEEL, STEERING COLUMN, AIRBAG, OR INSTRUMENT PANEL COMPONENT DIAGNOSIS OR SERVICE. DISCONNECT AND ISOLATE THE BATTERY NEGATIVE (GROUND) CABLE, THEN WAIT TWO MINUTES FOR THE SYSTEM CAPACITOR TO DISCHARGE BEFORE PERFORMING FURTHER DIAGNOSIS OR SERVICE. THIS IS THE ONLY SURE WAY TO DISABLE THE SUPPLEMENTAL RESTRAINT SYSTEM. FAILURE TO TAKE THE PROPER PRECAUTIONS COULD RESULT IN ACCIDENTAL AIRBAG DEPLOYMENT.

(1) Disconnect and isolate the battery negative cable. Remove the instrument cluster from the instrument panel. Disconnect the instrument panel wire harness connector (Connector C1) for the instrument cluster from the connector receptacle on the back of the cluster housing.

(2) Reconnect the battery negative cable. Turn the headlamps On and select the headlamp high beams with the left multi-function switch control stalk. Check for battery voltage at the high beam indicator driver circuit cavity of the instrument panel wire harness connector (Connector C1) for the instrument cluster. If OK, replace the faulty instrument cluster. If not OK, repair the open high beam indicator driver circuit between the instrument cluster and the left multi-function switch as required.

LOW FUEL INDICATOR

DESCRIPTION

Fig. 17 Low Fuel Indicator

A low fuel indicator is standard equipment on all instrument clusters (Fig. 17). The low fuel indicator is located near the lower edge of the instrument cluster, to the left of center. The low fuel indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “Fuel” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. An amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the icon to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The low fuel indicator is serviced as a unit with the instrument cluster.

OPERATION

The low fuel indicator gives an indication to the vehicle operator when the level of fuel in the fuel tank becomes low. This indicator is controlled by a transistor on the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The low fuel indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will only allow this indicator to operate when the instrument cluster receives a battery current input on the fused ignition switch output (run-start) circuit. Therefore, the LED will always be off when the ignition switch is in any position except On or Start. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the low fuel indicator for the following reasons:

- **Bulb Test** - Each time the ignition switch is turned to the On position the indicator is illuminated for about two seconds as a bulb test.
- **Less Than 12.5 Percent Tank Full Message** - Each time the cluster receives a message from the PCM indicating that the percent tank full is less than 12.5 (one-eighth), the low fuel indicator is illuminated and a single chime tone is sounded. The low fuel indicator remains illuminated until the cluster receives messages from the PCM for a continuous twenty seconds indicating that the percent tank full